Post-Disaster Rescue Guidance via VLM Fine-Tuning

CSE499A – Section 15, Group 5, Update 1

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Weekly Update:

Our primary objective this week was to architect and construct a highly robust, end-to-end data curation pipeline for our Vision Language Model (VLM). The goal was to programmatically ingest, normalize, and process raw, unstructured disaster imagery into a highly balanced, tactical instruction-response dataset ready for QLoRA fine-tuning.

Work Done by Abrar Mohammed Tanzim Alam

1. **Image Standardization Pipeline:** Built robust pre-processing scripts (`support/image_standardizer.py`) using the `PIL` library to optimize massive raw satellite and aerial imagery.
   - Enforces strict RGB color spaces; auto-detects and discards naturally grayscale images.
   - Caps maximum resolution at 1280px to prevent VRAM exhaustion (OOM errors) during Qwen2-VL training on T4 GPUs.
   - Filters out images with edge dimensions below 256px.
   - Tracks execution state to allow resume-safe processing.
   - Outputs standardized `.jpg` images and `.json` metadata labels into processed directories.
2. **Instruction-Response Ground Truth Generation:** Engineered the generation engine (`support/generate_ground_truth.py` and `support/aider_generate_ground_truth.py`) leveraging the `gemini 3.5 and 2.5 flash models` via google.genai SDK.
   - Implemented dynamic prompt multipliers to aggressively over-sample rare disaster events, addressing the class imbalances identified in EDA.
   - Yielded a highly balanced week 1 pilot batch of ∼500 pairs (496 exactly) formatted for tactical rescue guidance.
   - Output: `dataset/train_dataset.jsonl`.
3. **Dataset Sanitization Engine:** Created an automated sanitization script (`support/sanitize_dataset.py`) using regular expressions to rigorously clean both newly generated and legacy xView datasets.
   - Scrubs first-person conversational Ai language (e.g., "I apologize," "As an AI" etc) and strips generic placeholder brackets generated by the LLM.
   - Enforces strict length boundaries: discards responses under 80 words (truncated/insufficiently detailed) or over 600 words (hallucinatory/overly verbose).
   - Uses round-robin injection to replace repetitive default system prompts with 6 distinct tactical instruction variants, improving generalized query comprehension.
   - Output: `dataset/final_training_dataset.jsonl` – scrubbed, diverse, and randomized.
4. **Final Verified Dataset Statistics Visualization:** Developed `support/dataset_stats.py` to automatically parse the finetune-ready `dataset/final_training_dataset.jsonl` and generate comprehensive visual reports using `matplotlib` and `seaborn` libraries. This tool plots the post-generation distributions of disaster types and damage severities, providing immediate, verifiable proof that our balancing mechanisms successfully mitigated the raw dataset's intrinsic biases.
   - Prepared the small sample pool (496) for seamless ingestion into the proof-of-concept finetuning of the Qwen2-VL model from the xView dataset.
5. **Cross-Dataset Pipeline Adaptation (AIDER Transfer):** Successfully adapted the entire xView-centric data curation architecture to seamlessly ingest and process the AIDER aerial dataset. This required extensive refactoring of the standardizer and generation scripts to dynamically interpret AIDER's unique class-based directory structure and normalize differing coordinate systems without disrupting the core pipeline logic. So that the AIDER pipeline can utilize the downstream processing scripts of the xView dataset.

6. **Synthetic Metadata Synthesization:** Designed an advanced bridging script (support/aider_synthesize_metadata.py) specifically to solve AIDER's lack of precise ground-truth annotations. The script queries Gemini to synthetically generate xBD-equivalent metadata (bounding polygon approximations and structural severity classifications) and outputs them as temporary `.json` files, flawlessly normalizing the diverse AIDER data into our strict uniform training schema.

Joint Contribution: Tamanna Akter Mou, Aryan Sami, and Ridita Afrin Riya

1. **Reviewer Validation UI Tool:** Designed a local graphical user interface (support/dataset_validator.py) using `Tkinter` for rapid human-in-the-loop verification.
   - Ingests dataset/train_dataset.jsonl, rendering each disaster image alongside its synthesized response.
   - Allows Accept, Reject, or Undo evaluation against a strict Rescue Actionability Rubric (RAR), checking for:
     - Correct identification of structural collapses
     - Correct geospatial quadrant mapping
     - Hallucination-free hazard avoidance tips
   - Output: dataset/verified_dataset.jsonl (highest-quality samples only).

Work Done by Ridita Afrin Riya

1. **AIDER Script Adaptation:** Created the initial AIDER image standardization script by referencing and adapting the core architecture of the xView dataset standardization script (support/aider_image_standardizer.py).
2. **AIDER Dataset Processing:** Standardized and synthesized pseudo-labels for half (∼2700) of the raw AIDER dataset utilizing the core pipeline scripts.
3. **Data Generation:** Successfully generated an additional 50 ground-truth instruction-response pairs to from the AIDER dataset to expand our finetune-ready dataset.

Work Done by Aryan Sami

1. **Human-in-the-Loop Validation:** Acted as the primary reviewer utilizing the local GUI validator to manually review and evaluate 200 synthesized ground-truth instruction-response pairs against the strict Rescue Actionability Rubric (RAR).
2. **AIDER Dataset Processing:** Standardized and synthesized pseudo-labels for the remaining half (∼2700) of the raw AIDER dataset utilizing the core pipeline scripts.
3. **Data Generation:** Successfully generated an additional 50 ground-truth instruction-response pairs from the AIDER dataset to expand the training pool.

Work Done by Tamanna Akter Mou

1. **Datasets' Exploratory Data Analysis (EDA):** Developed comprehensive Jupyter notebooks (`xview_eda.ipynb`, `aider_eda.ipynb`) to programmatically analyze the raw dataset structures for both AIDER and xView datasets in the initial phases. Extracted critical statistical distributions regarding image resolutions, disaster categories, and damage severity levels. This foundational analysis exposed severe class imbalances which directly guided the architectural design of the downstream standardization and prompt augmentation pipelines.

Work Done by Abdullah Al Noman

1. **Human-in-the-Loop Validation:** Acted as a secondary reviewer utilizing the local GUI validator to manually review and evaluate the remaining 300 synthesized ground-truth instruction-response pairs from the xView dataset against the strict Rescue Actionability Rubric (RAR).
2. **Data Generation:** Successfully generated 50 ground-truth instruction-response pairs from the AIDER dataset to expand our finetune-ready dataset.

[*] **Next Week's Plan:** We will proceed to merge the fully processed datasets and begin the QLoRA fine-tuning phase using the prepared splits to train the Qwen2-VL model.